F78AB Actuarial and Financial Mathematics

Chapter 5: Real Interest Rates

5.1 The Changing Price of Sugar

Suppose I buy 1kg of sugar every year on 1 January.

How much does it cost me to buy 1kg of sugar?

How does the cost change from year to year?

Suppose I buy 1kg of sugar at the start of every month and suppose the prices I pay, in pence, each month are given in the table below.

5.1 The Changing Price of Sugar

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- How does the cost change from year to year?
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5.1 The Changing Price of Sugar

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
2000	166.6	167.5	168.4	170.1	170.7	171.1	170.5	170.5	171.7	171.6	172.1	172.2
2001	171.1	172.0	172.2	173.1	174.2	174.4	173.3	174.0	174.6	174.3	173.6	173.4
2002	173.3	173.8	174.5	175.7	176.2	176.2	175.9	176.4	177.6	177.9	178.2	178.5
2003	178.4	179.3	179.9	181.2	181.5	181.3	181.3	181.6	182.5	182.6	182.7	183.5
2004	183.1	183.8	184.6	185.7	186.5	186.8	186.8	187.4	188.1	188.6	189.0	189.9
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1

Table 5.1

Examples

Example 5.1.1

What is the annualised average change in the price of sugar from January 2000 to January 2005? **2.544%**

Example 5.1.2

With £16.66 in January 2000, I can buy 10kg of sugar.

With £16.66 in December 2005, how many kg of sugar can I buy? **8.58 kg**

Exercises

- In January 2000, I buy £100 nominal of a special 5-year bond, called a "Sugar Bond".
- The Sugar Bond pays no coupons but its redemption value is increased by the change in the price of sugar over the 5 years from January 2000 to January 2005.

Exercises

Exercise 5.1

Using Table 5.1, calculate the redemption payment at January 2005. **£113.385**

Exercise 5.2

Calculate how much sugar can be bought with

i. £100 in January 2000. **60.024 kg**

ii. the redemption payment in January 2005.

5.2 Price Inflation Indices

- Two important indices in the U.K. are
  - Retail Price Index (RPI).
  - Consumer Price Index (CPI).
- Measure the change in prices of a specified set (basket) of goods and services.
- Published by Office for National Statistics (ONS). [Department of Statistics Malaysia]
- Other countries use similar measures.

Exercises (for you to do)

Exercise 5.3

From the ONS website, find out how these two indices differ.

Exercise 5.2

Find out who makes use of RPI and CPI.

Example 5.2.1

- You have started your dream job. At the end of your first year, your employer offers you a pay rise of 1%.
- You hear in the news that RPI has increased by 4% over the year and is expected to stay high in the next few years.
- Would you be happy with your pay rise?

Example 5.2.2

- In 2012, you retired from your dream job. In 1962, you had deposited £100 000 in a savings account to provide for your retirement. The account has paid interest at an average effective rate of 6.14% per annum.

Example 5.2.2 cont.

- RPI has increased at an average effective rate of 5.85% per annum from 1962 to 2012. Thus, in terms of 1962 purchasing power, the accumulated 2012 deposit has a real value of
- In other words, £1 967 679 can buy as much in 2012 as £114 660 could buy in 1962.

Example 5.2.2 cont.

Real Returns

The average **real return** (or real yield) per annum on the deposit is

or

5.3 Retail Prices Index (RPI)

- The RPI measures the monthly change in the prices of a specified set of goods and services.
- It is published one month in arrears (usually in the middle of the following month).

5.3 Retail Prices Index (RPI)

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
2000	166.6	167.5	168.4	170.1	170.7	171.1	170.5	170.5	171.7	171.6	172.1	172.2
2001	171.1	172.0	172.2	173.1	174.2	174.4	173.3	174.0	174.6	174.3	173.6	173.4
2002	173.3	173.8	174.5	175.7	176.2	176.2	175.9	176.4	177.6	177.9	178.2	178.5
2003	178.4	179.3	179.9	181.2	181.5	181.3	181.3	181.6	182.5	182.6	182.7	183.5
2004	183.1	183.8	184.6	185.7	186.5	186.8	186.8	187.4	188.1	188.6	189.0	189.9
2005	188.9	189.6	190.5	191.6	192.0	192.2	192.2	192.6	193.1	193.3	193.6	194.1
2006	193.4	194.2	195.0	196.5	197.7	198.5	198.5	199.2	200.1	200.4	201.1	202.7
2007	201.6	203.1	204.4	205.4	206.2	207.3	206.1	207.3	208.0	208.9	209.7	210.9
2008	209.8	211.4	212.1	214.0	215.1	216.8	216.5	217.2	218.4	217.7	216.0	212.9
2009	210.1	211.4	211.3	211.5	212.8	213.4	213.4	214.4	215.3	216.0	216.6	218.0
2010	217.9	219.2	220.7	222.8	223.6	224.1	223.6	224.5	225.3	225.8	226.8	228.4
2011	229.0	231.3	232.5	234.4	235.2	235.2	234.7	236.1	237.9	238.0	238.5	239.4
2012	238.0	239.9	240.8	242.5	242.4	241.8	242.1	243.0	244.2	245.6	245.6	246.8
2013	245.8	247.6	248.7	249.5	250.0	249.7	249.7	251.0	251.9	251.9	252.1	253.4
2014	252.6	254.2	254.8	255.7	255.9	256.3	256.0	257.0	257.6	257.7	257.1	257.5

Table 5.2

Example 5.3.1

The change in RPI from October 2012 to October 2013 is

Meaning the prices of the basket of goods and services on which the RPI is based has increased by 2.565% over the one year.

Example 5.3.2

The change in RPI from October 2011 to October 2012 is

Meaning the prices of the basket of goods and services on which the RPI is based has increased by 3.193% over the one year.

Example 5.3.2

The annualised average change in RPI from October 2011 to October 2013 is

Meaning the prices of the basket of goods and services on which the RPI is based has increased at an average effective rate of 2.879% per annum.

Exercise 5.5

Calculate the annualised average change in RPI from January 2005 to November 2013.

Exercise 5.6

Does the RPI always increase?

No.

- June 2001 to January 2002 values.
- May – July 2013 values.

Example 5.3.4 (for you to do)

- In the 2010 UK Budget, the Government announced that "The Government will adopt the CPI for the indexation of benefits, tax credits and public service pensions from April 2011" (Source: HM Treasury, Budget 2010, page 17, paragraph 1.43). Previously, RPI had been used.
  - What reason did the Government give for the change?
  - Is there an impact on the recipients of the benefits, tax credits and pensions?

Next Chapter

Chapter 6: Force of Interest and Continuously-paid Cashflows